

MARKOVO, Russian Federation

WMO: 255510

Lat: 64.6742N

Lon: 170.4278E

Elev: 26

StdP: 101.01

Time Zone: 12.00 (E12)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-46.5	-44.3	-48.9	0.0	-46.4	-46.8	0.0	-44.3	8.5	-10.6	7.1	-11.3	0.3	230	0.765

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	25.7	16.5	23.6	15.6	21.5	14.8	17.8	23.8	16.5	21.8	15.4	20.1	2.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.4	11.0	19.9	14.1	10.1	18.8	13.0	9.4	17.5	50.4	23.9	46.6	21.8	43.2	20.1	23.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.0	5.9	5.1	DB	-47.4	28.8	2.6	1.9	-49.2	30.2	-50.7	31.3	-52.2	32.4	-54.1	33.8	(4)
			WB	-47.4	19.6	2.8	1.5	-49.4	20.7	-51.0	21.6	-52.6	22.5	-54.6	23.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-7.4	-26.8	-25.1	-20.4	-13.1	0.5	11.6	14.7	11.5	5.0	-6.4	-16.8	-24.1	(6)
(7)		DBStd	16.67	10.21	9.81	8.23	7.15	5.19	3.93	3.43	3.34	4.09	6.53	9.19	11.19	(7)
(8)		HDD10.0	6636	1142	984	944	692	296	28	4	23	156	507	803	1057	(8)
(9)		HDD18.3	9384	1400	1217	1202	942	554	203	120	212	400	766	1053	1315	(9)
(10)		CDD10.0	302	0	0	0	0	1	76	149	70	6	0	0	0	(10)
(11)		CDD18.3	9	0	0	0	0	0	1	7	1	0	0	0	0	(11)
(12)		CDH23.3	195	0	0	0	0	0	53	130	12	0	0	0	0	(12)
(13)		CDH26.7	26	0	0	0	0	0	5	21	1	0	0	0	0	(13)
(14)	Wind	WSAvg	2.1	1.8	2.1	2.2	2.2	2.4	2.1	2.0	2.1	2.1	2.1	2.1	(14)	
(15)	Precipitation	PrecAvg	399	23	22	24	18	21	29	41	59	44	37	46	35	(15)
(16)		PrecMax	522	50	63	75	38	45	64	105	105	109	140	77	83	(16)
(17)		PrecMin	291	0	2	1	1	6	5	4	24	2	5	15	1	(17)
(18)		PrecStd	79	14	17	15	11	11	16	25	24	28	26	18	20	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.2	-1.1	2.4	3.4	16.7	27.3	29.2	25.0	20.0	8.2	2.1	1.4	(19)
(20)		MCWB	-0.7	-1.7	0.8	0.7	9.2	15.2	18.5	17.0	13.5	6.4	1.0	0.4	(20)	
(21)		2%	DB	-4.2	-3.8	-1.1	1.3	13.1	25.0	26.5	22.3	16.6	5.7	0.6	-1.7	(21)
(22)		MCWB	-4.8	-4.1	-2.1	-0.4	7.2	14.6	17.5	15.2	10.8	4.3	0.1	-2.3	(22)	
(23)		5%	DB	-8.0	-6.7	-4.2	-0.1	10.6	22.9	24.6	20.1	14.0	3.7	-1.2	-4.8	(23)
(24)		MCWB	-8.4	-7.1	-4.9	-1.3	6.1	13.9	16.7	14.5	9.9	2.4	-1.5	-5.2	(24)	
(25)		10%	DB	-11.8	-9.9	-8.2	-2.2	7.9	20.5	22.6	18.1	11.9	1.5	-3.9	-8.1	(25)
(26)		MCWB	-12.2	-10.3	-8.7	-3.5	4.5	13.0	16.1	13.7	8.9	0.3	-4.4	-8.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.8	-1.6	1.0	1.0	9.6	17.3	19.7	18.4	14.0	6.9	1.2	0.3	(27)
(28)		MCDWB	0.2	-1.1	2.3	2.6	15.8	25.0	25.7	23.3	18.4	8.0	1.8	1.3	(28)	
(29)		2%	WB	-4.7	-4.1	-2.1	-0.1	7.7	15.3	18.6	16.3	11.7	4.6	0.1	-2.2	(29)
(30)		MCDWB	-4.2	-3.8	-1.2	1.1	12.8	22.6	24.8	20.3	14.9	5.6	0.4	-1.6	(30)	
(31)		5%	WB	-8.4	-7.1	-5.0	-1.3	6.2	14.2	17.5	15.1	10.5	2.6	-1.5	-5.3	(31)
(32)		MCDWB	-8.0	-6.7	-4.2	-0.1	10.3	21.6	23.4	19.1	13.0	3.4	-1.1	-4.8	(32)	
(33)		10%	WB	-12.2	-10.3	-8.7	-3.4	4.7	13.3	16.4	14.0	9.3	0.5	-4.3	-8.6	(33)
(34)		MCDWB	-11.8	-9.9	-8.2	-2.2	7.8	20.1	21.6	17.4	11.5	1.3	-4.0	-8.2	(34)	
(35)	Mean Daily Temperature Range	MDBR	8.8	10.1	11.2	11.1	8.2	10.7	10.7	10.0	9.2	7.6	8.2	8.3	(35)	
(36)		5% DB	MCDBR	9.2	8.2	9.1	8.0	12.5	16.7	16.0	14.6	14.6	6.7	6.5	8.4	(36)
(37)		MCWBR	9.1	8.1	8.3	6.5	7.3	7.6	8.3	8.4	9.5	5.2	6.3	8.2	(37)	
(38)		5% WB	MCDBR	9.3	8.0	9.1	7.5	11.6	14.4	14.0	12.2	11.4	5.4	6.5	8.3	(38)
(39)		MCWBR	9.2	7.9	8.3	6.0	7.0	7.1	7.7	7.4	7.7	4.4	6.4	8.1	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.166	0.240	0.303	0.358	0.410	0.394	0.502	0.486	0.339	0.277	0.189	0.175	(40)	
(41)		taud	1.938	2.145	2.064	1.951	1.998	2.298	1.953	1.946	2.398	2.253	1.929	1.925	(41)	
(42)		Ebn at Noon	407	676	758	790	785	819	695	656	737	612	326	126	(42)	
(43)		Edh at Noon	24	57	99	141	149	114	157	143	73	51	23	9	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.14	0.85	2.62	4.86	5.64	5.51	4.94	3.69	2.21	0.93	0.24	0.05	(44)	
(45)		RadStd	0.01	0.09	0.23	0.20	0.50	0.37	0.50	0.38	0.23	0.11	0.03	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+1.14	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page